

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (EC)

May 2013

EC 710 Digital Image and Speech Processing

Date: 15.05.2013, Wednesday

Time: 10:00 p.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION-I

- Q-1 a) Define: (i) Formants (ii) Vowel Triangle (iii) Vocal Tract 3
- b) Find the inverse z-transform of 4

$$X(z) = \frac{z+3}{z^7(z-0.5)}, \text{ where } x(n) \text{ is causal sequence.}$$

- c) Describe applications of digital speech processing. 4
- Q-2 a) Explain: Pitch period estimation using the Autocorrelation Function. 8
- b) Derive the vocal tract system function using two concatenated lossless tubes. 4

OR

- a) The short-time energy of a sequence $x(n)$ is defined as 8

$$E_n = \sum_{m=-\infty}^{\infty} [x(m)w(n-m)]^2$$

For the particular choice

$$w(m) = \begin{cases} a^m & m \geq 0 \\ 0 & m < 0 \end{cases}$$

It is possible to find a recurrence formula for E_n .

- (i) Find a difference equation that expresses E_n in terms of E_{n-1} and the input $x(n)$.
- (ii) Draw a digital network diagram of this equation.
- (iii) What general property must

$$h(m) = w^2(m)$$

have in order that it be possible to find a recursive implementation?

- b) Explain Flanagan speech production mechanism. Draw general discrete time model for speech production. 4

- Q-3 a) What is instantaneous Quantization? Describe comparison between two uniform quantizer characteristics. 8
- b) Explain Durbin's recursive solution of the autocorrelation equation for order 2 using predictor coefficients equation. 4
- OR
- a) Describe the concept of differential quantization in delta modulation. Compare it with Adaptive delta modulation. 8
- b) Write short note on: Overlap addition method of short-time synthesis. 4

SECTION-II

- Q-4 a) What is digital image processing? List various fields of digital image processing and necessary tools for processing. 4
- b) Describe image formation in the human eye. Correlate it with the image sensing and acquisition system. 4
- c) Explain the concept of gray level slicing with example. 3
- Q-5 a) Explain: Significance of histogram of an image in terms of probability. What would be the effect on the histogram of given image P if we set the higher order bit planes to zero? 4

P =

0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

Comment on resultant image.

- b) What is the basic difference between image restoration and image enhancement process? Draw complete block diagram of Image restoration process and explain each block. 8
- OR
- a) Filter the following image Q by using a 3 x 3 neighborhood averaging by assuming pixel replication. Give the complete numerical solution. 4

Q =

1	2	3	2
4	2	5	1
1	2	6	3
2	6	4	7

- b) What will be the value of the center pixels change to when the given image Q (As given in Que 5(b)) is passed through a
- (i) Arithmetic mean filter 8

(ii) Geometric mean filter

(iii) Harmonic mean filter

(iv) Max filter

Q-6 a) What are the advantages of processing an image in the frequency domain? What is the correspondence between smoothing and sharpening filtering respectively, in the spatial and frequency domain? 8

b) Express: Pseudo-color image processing. 4

OR

a) Explain: The Laplacian filtering in frequency domain. Compare it with other frequency domain filters. 8

b) Write short not on: Color Transformations 4
